UCLA Miniscope Integrated Baseplate + GRIN

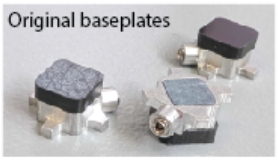
Written By: Dr. Sandra Poulson, Corder Lab at Penn

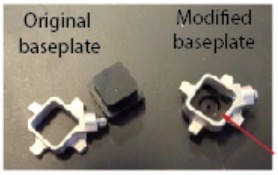
This can allow you to do a small craniotomy to inject virus, then do the lens implant after surgery to ensure locating your cells with the miniscope. *Note: for any of the supplies and tools listed here, you can use your discretion to buy from an appropriate supplier. e.g. your university may have different suppliers for glues, etc.*

Notes from 20 Feb 2025 Visiting Daiki from the Max Kelz lab on UCLA miniscope GRIN implants:

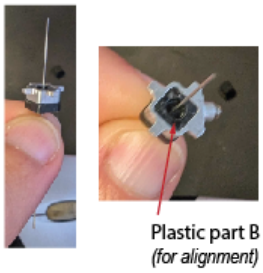
(Pack of Five) V4 Variant 2 Base Plate three piece set w/2-56 set screw tapered point & cap from miniscope parts https://miniscopeparts.com/products/ols/products/v4-variant-2-base-plate-w-pointed-set-screw

These are original baseplate (as sent by manufacturer) and modified baseplate (as by UPenn instrumentation shop):

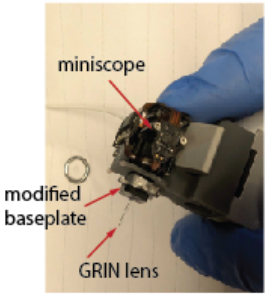




These is how the plastic part and the metal parts of the baseplate are aligned to be glued together:



This is how the miniscope and the modified baseplate with the GRIN lens attach:

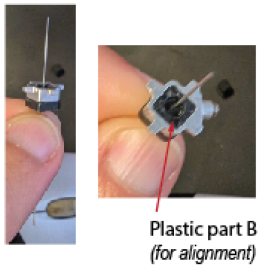


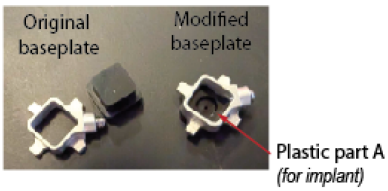
This is how the modified baseplate is implanted:



This is how the modified baseplate sits on the head of the animal:

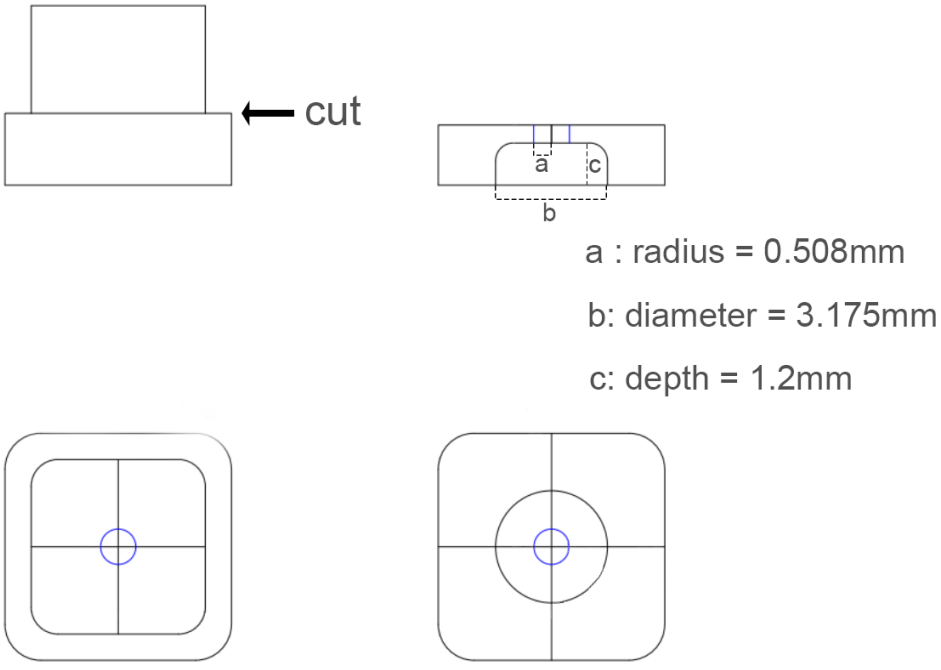






In these first two photos, Daiki is showing an early iteration here where he had the hole drilled through the entire cap. What you want is to cut the to part of the cap off, so that it's just the wider part. That's the part that gets glued to the "bottom" of the metal baseplate in this third photo. You can see that the cylindrical shape is inside, and you set the miniscope inside the baseplate on recording days. This gives a small but repeatable gap between the miniscope and the GRIN, which the Kelz/Proekt labs found is ideal for imaging.

Note as of October 2025: Shylo Stiteler of miniscopeparts.com can modify his baseplates such that you can just order and use. If you mention that you want to use these same dimensions (good for 1mm GRIN), you can mention the design he created for me. If you need this for a different GRIN, you can work with Shylo to modify the diameter or the depth. Shylo sells in pack of 5 and 10 https://miniscopeparts.com/products/ols/products/pack-of-five-v4-variant-2-base-plate-three-piece-set-w-slash-2-56-set-screw-tapered-point-and-cap





This is the product you can buy directly from Miniscopeparts.com. Shylo will send 1) the modified cut cap with the hole (bottom row), 2) the intact cap that is used as a mold when gluing the metal baseplate to the cut cap (top row), and 3) the metal baseplate frame with the screw.

This was the instruction for a 0.5mm GRIN:

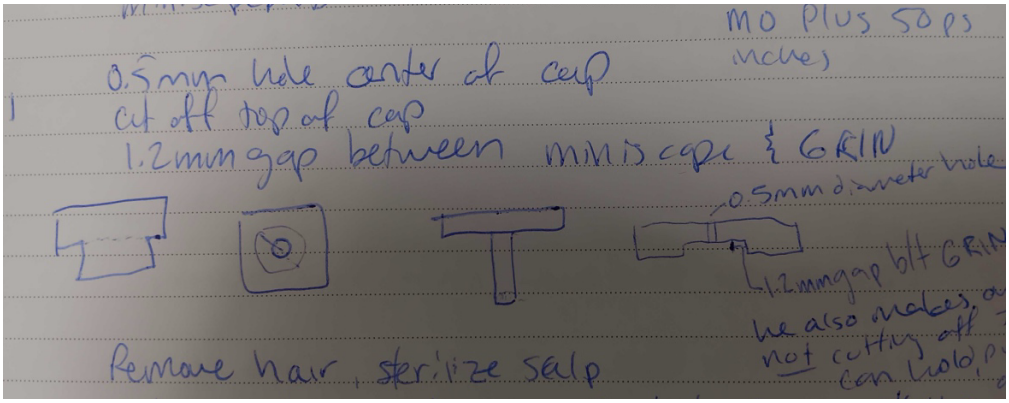
0.5mm diameter hole in center of cap

Cut off top of cap

1.2 mm gap depth between miniscope & GRIN

This is Part A

Part B is the same thing but without cutting off the base of the cap so that he can hold it and put the rod in the center and glue the modified cap to the metal baseplate

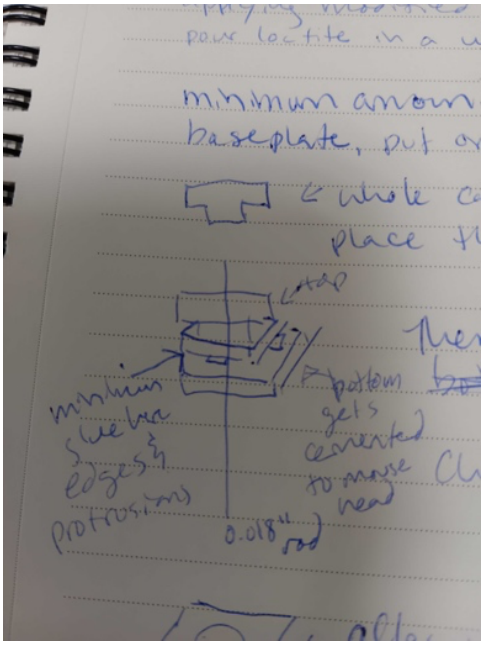


Align the whole cap, the cut cap, and the metal baseplate on a steel gauge pin at the appropriate size for your GRIN. Example steel gauge pin set https://www.amazon.com/Storage-Tolerance-Calibrates-Measures-Deviation/dp/B0CZ6G3S28

The cylindrical indention goes on the inside of the metal baseplate to allow you to position the miniscope on top with a slight gap between the GRIN and the miniscope lens. This has been tested by the Kelz lab, but if you feel this gap is not what you need, you may need to modify the depth.

To glue the modified headcap to the baseplate

1) Pour Loctite in a weigh boat
2) Put a minimal amount of glue around bottom edge & corners along the interface between the plastic cut cap and metal baseplate frame, allow to dry



Then remove the Part B whole cap
In this drawing, the whole cap has 0.5mm hole in the middle to place the rods through the setup, rod is 0.018" (~0.457mm)

3) After glue is dry, disassemble the rod setup. Then, carefully insert the GRIN lens through the hole & push until flush with the inner edge of the center hole.

4) Check FOV with miniscope. The integrated baseplate + GRIN is now ready for implant. You can try injecting virus first, allowing expression, then implanting this integrated baseplate with miniscope attached to visualize cells prior to laying down cement, or you can inject virus in the same surgery as implant and wait 2 weeks before recording for viral expression.

5) After implant (if virus was already expressed), wait at least 1 weeks prior to recordings. You can try a cap, but be aware that any plastic the mice will chew. You can try Kwik-sil (see below) in the hole to prevent dust. It honestly may save time, but the method used in the Corder lab with integrated baseplate + GRIN is to flood the area with ethanol, let sit for ~10 min, soak out with the twisted tip of a Kim wipe, repeat as necessary to eliminate dust. This honestly is time consuming so if you cover with Kwik-sil and remove on recording days, this might result in saved time.

Detailed lens implant instructions after use 300nl virus expressed for at least 2 weeks

1) Anesthetize animal, apply eye cream (ophtHAvet 0.4% sodium hyaluronate) to keep moist, headfix, remove hair, sterilize scalp

2) Make incision and cut away scalp to expose skull surface

3) Score skull (cross-hatch) and level

4) Using 0.5mm burr, make a hole just at the spot where GRIN will go

5) Also make a burr hole for anchor screw – might take a bigger burr (Daiki also makes 2 holes for eeg and ground)

6) Use a larger bur to soften edge of bone around holes

7) Check head leveling after placing screws

8) Using a SANANTS blunt needle @ 0.5mm (24ga)

https://www.amazon.com/Unsterilized-Synthetical-Dispensing-Needle-Blunt/dp/B07Z7KKY2K/

attached to a vacuum line, slowly lower vacuum + needle (Daiki stops ~1.5DV). Start vacuum, use for short period, repeat as needed (e.g. 2-3 times) to ensure tissue and blood removed. Shorter is better to prevent continued bleeding. *Note from the Corder lab: if you are not implanting deeply, e.g. many superficial cortical regions, you may not need vacuum tissue removal. In this case, implant the GRIN similarly to optic fiber. However, you NEED to remove the entire dura. The dura looks like a bumpy textured surface and it takes a careful eye to recognize it. Remove with extremely sharp, pointed forceps and remove all of it or it will obscure your FOV. When you successfully remove the dura, the underlying cells will be very very shiny, not a bumpy texture. Also note: you want to minimize bleeding altogether for calcium imaging. When blood gets between your GRIN and your cells, you will not get good calcium signal. So soak any blood with Curaspon or Vetspon (Curaspon is the human dental version and is often cheaper), keep flushing, keep things moist, and watch for bleeding to stop prior to inserting your GRIN.*

9) Use miniscope software to visualize location of GRIN as you lower into place

: LED intensity

: Focal plane

: Frame rate

: Gain

: Upper level of displayed frame. Does not affect saved data.

: Lower level of displayed frame. Does not affect saved data.

Daiki was using between 10 and 37 for LED, ~50 for focal plane, 15-30FPS, G=1

10) To lock baseplate in place:

Daiki used this method to adhere baseplate to skull:

Loctite 495

Add Insta-set Bob Smith Industries Accelerator

TEETS coral denture material and methyl methacrylate w/ TEETS “cold cure” liquid compound

Note – While using cyanoacrylates are common to try to get dental cement to adhere to bone, cyanoacrylates can cause embolization when entering the blood stream and polymerizing within vasculature, so keep this fully away from bleeding areas. If you can use metabond instead, do, because it lowers this risk.

Notes from 11 Feb 2025 visiting Ian Chapman of Kevin Bolding’s lab at Monell:

This method expresses a GCaMP virus and installs the GRIN in one surgery (separate step done 4-6 weeks before baseplate installation, similar to UCLA miniscope workshop instructions).

Steps for cutting away scalp and cross hatching are the same

Skull should be VERY dry, keep dura soft with saline

After injecting virus (50-150ul), ease lens in 1-2mm at a time

Use bob smith industries insta-cure (purple) and put a drop into burr hole

Use insta-weld activator 2303-6 to dry if needed but don't get into hole

Apply metabond with toothpicks

Using metabond green tub, mix 1 level scoop, 4 drops of base, mix 1 drop catalyst

Focus metabond on covering the middle base layer for cap

For edges use Insta-cure blue bottle (super thin), pour on using toothpick
https://www.amazon.com/BobSmithIndustries-Insta-Cure-Super-Thin-Glue/dp/B07FN8KQZQ

Apply black dental cement on surface, build up around edges

Apply Kwik-sil (equal parts) to lens to cover https://wpiinc.com/products/kwik-sil-kwik-cast-silicone-bioadhesives-and-sealants?srsltid=AfmBOooU63hXZ9iDLCmasqixUpEQ0qKyoYxO0iCRHSkJA0JeOHDLHb3t

Apply layer of white dental cement over Kwik-sil, will keep like this until ready to apply baseplate.

When ready to remove, drill around edges to remove white dental cement, Kwik-sil should be soft underneath, clean up grit with ethanol if needed.

Baseplate Session (2+ weeks later) with Ian 6 May 2025 visit:

1) Drill around the edges to remove the white dental cement, Kwik-sil should be soft underneath and can pluck off
2) Remove Kwik-sil

    Osada exl-m40 dental drill

3) Slowly lower scope while software is on
4) Using software -> 2 lightbulb

    ->10 - 17 up down arrow this is digital depth

    ->30 timer

    ->3.5 Gain

    ->1.00 A

    ->0.00 B

5) In this case might need to drill cement to allow miniscope to get close enough, clear off lens with ethanol
6) Place baseplate where cells may be present, make sure a nice focal range on both ends exists. His is 0.5 mm lens so won't fill the FOV of the miniscope lens. Apply stimulus (in this case odorant) to ensure visualization of cells. Other options for stimulus: foot pinch, drug

This kind of tool can give you maximum precison of glue placement https://www.grainger.com/product/MOODY-TOOL-Precision-Pick-Steel-6KHL0
Purple glue Bob Smith industries Purple Bob Smith Industries glue https://www.amazon.com/Insta-cure-filling-Bob-Smith-Ind/dp/B0000DD1QP/ and - Polyvance 2303-6 insta-weld activator from your favorite hardware supplier. Apply drops of the purple glue under the baseplate feet, then roll a drop of the insta-weld to lock in place with the precision tool. Add a couple more dollops of the glue and lock in place before moving the scope, then lock baseplate into place. Loosen ear bars prior to removing scope to not stress glue. Apply drops of purple glue

under metal “feet” protrusions of baseplate with an angled probe, then roll drop of insta-weld to lock in place. Move miniscope out of the way for rest of procedure.

7) If using a head bar for head fixed recording, cement this in now.
8) Then he builds up the black dental cement around the corners first, then around the edge
9) He uses a narrow sharp scalpel blade to remove excess. I’m listing some options here in case they help.
   -https://www.emsdiasum.com/disposable-scalpel-mini-11-20pk?srsltid=AfmBOoorE9psyzn1NseHRWAygNP-UG3PIx2XgNSB3sLp5FiigkYPeDfpsuE
   -https://unitedsupplycorp.com/products/mini-scalpel-disposable-11?pr_prod_strat=e5_desc&pr_rec_id=363753ba2&pr_rec_pid=9533865263381&pr_ref_pid=9533865165077&pr_seq=uniform
   -https://mymedicaloutlet.com/products/graham-field-feather-sterile-microscalpels
10) Allow at least 1 week recovery prior to recordings.

Option for UCLA scope holder:
https://optogeneticsandneuralengineeringcore.gitlab.io/ONECoreSite/projects/Miniscope/#miniscope-holders note this is for V3 so must be modified to fit V4

Notes from Blake on Inscopix Integrated GRIN+Baseplate Surgery:

Miniscope Implants

Inject virus 2 week prior to miniscope surgery.

Set up either the Inscopix or UCLA DAQ box and laptop prior to starting the surgery – will need this to visualize fluorescent neurons during lens insertion.

Have the miniscope attached to the stereotax arm adaptor

DV for the lens will be zeroed at the surface of the brain for ACC, while for PAG is preferable to zero as close to bregma as possible without touching the bone.

1) Remove scalp – can just cut out circular piece of skin to make room for the lens/baseplate and dental cement
2) Apply 30% hydrogen peroxide to the skull with a qtip and completely dry (skull will appear different after. The sutures will be very visible)
3) Use a smaller drill bit and drill out the hole in a small square-shaped pattern to accommodate the lens; don’t break through dura with the drill bit
4) Make additional divots to accommodate at least one skull screw
5) After drilling, remove the right stereotax arm to make more room for working with the skull/lens/baseplate
6) Use sharp pointed forceps to remove the dura
7) Continuously monitor drill hole for bleeding – must not be bleeding during lens insertion
8) Prepping the baseplate/lens and inserting in brain
   a. There is 1 screwdriver per miniscope that stays in the Inscopix box; this is needed to loosen the screw on the baseplate
   b. Magnetically attach the baseplate to the miniscope; tighten the set screw (do not overtighten)

    c. Attach to left stereotax arm and position the lens over Bregma (center of the lens is used to zero A/P and M/L)
    d. Turn off lamp and have miniscope on to help with visualization
    e. Move lens to the coordinates and zero D/V at the brain surface for ACC (if bleeding occurs, gently clean with lens paper and stop bleeding in the drill hole)
    f. Slowly lower the lens to the correct D/V coordinate
9) Apply metabond to cover the entire skull and build up around the lens – metabond should physically connect the skull, skull screws, lens, and bottom of the baseplate – do not get metabond on the miniscope
    a. Reminder for how to use metabond:
        i. 1 scoop green jar, leveled
        ii. 3 drops liquid dropper
        iii. Mix with wood stick
        iv. Rotate catalyst (few turns) until 1 drop comes out of syringe
        v. Pu on surface of skull and around lens
        vi. Wait 1-2 mins
10) Add dental cement (can add iron oxide if desired) – again, avoid touching the miniscope and do not cover the set screw on the baseplate
11) Once cement is set, loosen the set screw and, while holding the baseplate with curved forceps, raise the stereotax arm to disconnect the miniscope from the baseplate